

CALRIG · DESIGN DECISION RECORD

CalRig TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	CLR-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 (collocation partner) decided by Amish on 2026-10-02
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review recommendations adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** items A1 to A6 decided by Amish, 2026-09-25: go with recommendation (see CLR-DDR-002). Item O1 decided by Amish, 2026-10-02 (CLR-DEC-001).

■ Context

The TRL 2 review note (`docs/REVIEW.md` , session 2026-09-25, /populate) listed seven items as "Proposed, awaiting Amish". On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He did not review this batch item by item. Every item that carried a recommendation is therefore adopted as recommended for TRL 3 work, open for his review. Items without a recommendation stay open. On 2026-09-25 Amish then accepted all the recommendations ("i accept all your recommendations, go with them across all repos"), so A1 to A6 are now decided; see CLR-DDR-002.

■ Options considered

The options for each item are those listed in `docs/REVIEW.md` (session 2026-09-25, /populate) and in CLR-PRC-001 v0.2, Open questions.

■ Decision

Table 1. Items decided.

#	Item	Recommendation adopted	Status
A1	Budget	Option (a): raise <code>budget_usd</code> from \$300 to \$400. Set in <code>project.yaml</code> on 2026-09-25 (CLR-DDR-002).	Decided by Amish, 2026-09-25: go with recommendation.
A2	Reference strategy	Transfer references (two SHT45, SCD30, collocated SPS30) checked against salt fixed points, an ice point and field collocation, with the uncertainty chain stated in every report.	Decided by Amish, 2026-09-25: go with recommendation.
A3	Cooling limit (R1)	Accept 10 °C only in rooms at 25 °C or below for the first build; no second module or ice-water exchanger. R1 is redefined accordingly in CLR-REQ-001 v0.3.	Decided by Amish, 2026-09-25: go with recommendation.
A4	Test aerosol	Incense smoke decay; every report states the aerosol type.	Decided by Amish, 2026-09-25: go with recommendation.
A5	Gas scope	NO2 by field collocation only for now; no gas cylinder, dilution or toxic gas handling in CalRig. AirStreet's README already states that NO2 is calibrated by field collocation; its pitch line still says "calibrated on CalRig" and is AirStreet's to change. AirStreet is not edited from this repo.	Decided by Amish, 2026-09-25: go with recommendation.
A6	Door condensation	Insulated removable door panel for hot, humid points (and for cold points).	Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Items that remained open.

#	Item	Status
O1	Collocation partner for the transfer SPS30 (regulatory monitoring station, university site or an AQ-SPEC style program). No recommendation was made.	Decided by Amish, 2026-10-02, as recommended in CLR-DEC-001: a state or local regulatory monitoring site with a federal equivalent PM2.5 monitor; first candidate to approach a Texas Commission on Environmental Quality site in Dallas-Fort Worth, with South Coast AQMD's AQ-SPEC program as the fallback (CLR-DEC-001)

The pitch and problem lines in `project.yaml` were not flagged for rewording in the review and are unchanged.

■ Consequences

- CLR-REQ-001 v0.3: R1 redefined to 10 to 40 °C in rooms of 15 to 25 °C, with the lowest reachable set point stated in each record for rooms up to 30 °C (A3). R12 keeps the \$300 target from `project.yaml` and notes the proposed \$400.
- CLR-PRC-001 v0.3: the key design choices above are no longer shown as "proposed"; they are adopted for TRL 3 work, open for review. The door panel becomes part of the jacket (A6).
- CLR-PRB-001 v0.3: the constraint and out-of-scope lines reflect A1 and A5.
- CLR-CAL-001 v0.1 checks every requirement against the adopted design. It found that R2 (20 °C at 85 % RH in the reference room), R8 (O1 still open), R12 (cost) and R13 (mass) are not met. The design changes it made (full jacket, 120 mm mixing fan, sleeved bubbler with a trace-heated line, 600 x 500 mm base) are engineering changes within the adopted concept. The larger inner sink and the other TRL 3 review items were decided on 2026-09-25 and are recorded in CLR-DDR-002, which also brings R2 and R13 to met.